

Week 2 #2

1.5 Summations and Generalizations

If we have to represent many variables, rather than introduce different variables, it is convenient to represent them using subscripts: a_1 for first variable, a_2 for the second, etc, a_k for k-th variable.

Suppose you have n integers, say $a_1, a_2, a_3, \dots, a_n$. We refer to them as a *series* of integers. The sum of this series of integers is

$$S = a_1 + a_2 + a_3 + \dots + a_n$$

It is customary and convenient to use a summation notation, sigma, to represent this sum:

$$S = \sum_{k=1}^{k=n} a_k$$

k is called the index of the summation. The term a_k is a general term in the series. $k = 1$ is the lower limit of the sum, and $k = n$ is the upper limit of the sum. If the context makes the index variable clear, we can even write:

$$S = \sum_1^n a_k$$

You can generalize the use of summation notation to other operations.

For example, for the union of four sets, A_1, A_2, A_3, A_4 , we use the notation

$$A_1 \cup A_2 \cup A_3 \cup A_4 = \bigcup_{i=1}^4 A_i$$

Definition 1.5.3 Generalized Set Operations. Let $A_1, A_2, \dots, A_n$ be sets. Then:

$$(a) \quad A_1 \cap A_2 \cap \dots \cap A_n = \bigcap_{i=1}^n A_i$$

$$(b) \quad A_1 \cup A_2 \cup \dots \cup A_n = \bigcup_{i=1}^n A_i$$

$$(c) \quad A_1 \times A_2 \times \dots \times A_n = \prod_{i=1}^n A_i$$

$$(d) \quad A_1 \oplus A_2 \oplus \dots \oplus A_n = \bigoplus_{i=1}^n A_i$$

1.5.3 Exercises

1. Calculate the following series:

(a) $\sum_{i=1}^3 (2 + 3i)$

(c) $\sum_{j=0}^n 2^j$ for $n = 1, 2, 3, 4$

(b) $\sum_{i=-2}^1 i^2$

(d) $\sum_{k=1}^n (2k - 1)$ for $n = 1, 2, 3, 4$

2. Calculate the following series:

(a) $\sum_{k=1}^3 k^n$ for $n = 1, 2, 3, 4$

(b) $\sum_{i=1}^5 20$

(c) $\sum_{j=0}^3 (n^j + 1)$ for $n = 1, 2, 3, 4$

(d) $\sum_{k=-n}^n k$ for $n = 1, 2, 3, 4$

- 3.

(a) Express the formula $\sum_{i=1}^n \frac{1}{i(i+1)} = \frac{n}{n+1}$ without using summation notation.

(b) Verify this formula for $n = 3$.

(c) Repeat parts (a) and (b) for $\sum_{i=1}^n i^3 = \frac{n^2(n+1)^2}{4}$

4. Verify the following properties for $n = 3$.

(a) $\sum_{i=1}^n (a_i + b_i) = \sum_{i=1}^n a_i + \sum_{i=1}^n b_i$

7. For any positive integer k , let $A_k = \{x \in \mathbb{Q} : k - 1 < x \leq k\}$ and $B_k = \{x \in \mathbb{Q} : -k < x < k\}$. What are the following sets?

(a) $\bigcup_{i=1}^5 A_i$

(c) $\bigcap_{i=1}^5 A_i$

9. The symbol $\prod$ is used for the product of numbers in the same way that $\sum$ is used for sums. For example, $\prod_{i=1}^5 x_i = x_1 x_2 x_3 x_4 x_5$. Evaluate the following:

(a) $\prod_{i=1}^3 i^2$

(b) $\prod_{i=1}^3 (2i + 1)$

2.1 Basic Counting Techniques - The Rule of Products

If two operations must be performed, and if the first operation can always be performed p_1 different ways and the second operation can always be performed p_2 different ways, then there are $p_1 * p_2$ different ways that the two operations can be performed.

An extension of the above principle is:

The rule of products can be extended to include sequences of more than two operations. If n operations must be performed, and the number of options for each operation is $p_1, p_2, \dots, p_n$ respectively, with each p_i independent of previous choices, then the n operations can be performed $p_1 p_2 \dots p_n$ different ways.

This assumes that we can perform each operation independently of how the other is done.

If you have 4 ways to drive to Kingston from New Paltz, and 5 ways to drive from Kingston to Albany, how many different ways are there to drive from New Paltz to Albany?

2.2 Permutations

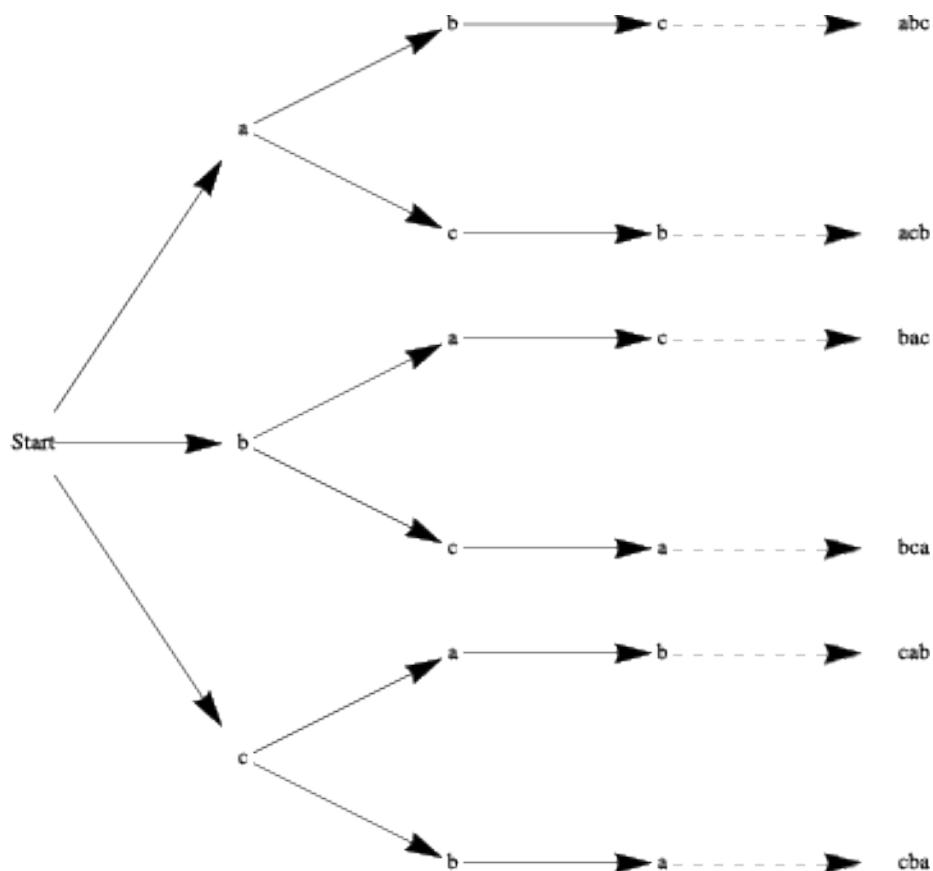
2.2.1 Ordering Things

A number of applications of the rule of products are of a specific type, and because of their frequent appearance they are given their own designation, permutations.

Consider the following examples.

Example: How many different ways can we order the three different elements of the set $\{a, b, c\}$?

Since we have three choices for position one, two choices for position two, and one choice for the third position, we have, by the rule of products, different ways of ordering the three letters. We illustrate through a tree diagram.



Each of the six orderings is called a **permutation** of the set. Note that there are no repetitions in a permutation of a set of objects.

Example: A student is taking five courses in the fall semester. How many different ways can the five courses be listed?

The first item in the list can be any one of the 5 courses. The second position in the list can list any one of the remaining 4 courses, and so on.

There are $5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$ different permutations of the set of courses.

In each of the above examples of the rule of products we observe that:

- a. We are asked to order or arrange elements from a single set.
- b. Each element is listed exactly once in each list (permutation). So if there are n choices for position one in a list, there are $n - 1$ choices for position two, $n - 2$ choices for position three, etc.

Example: The alphabetical ordering of the players of a baseball team is one permutation of the set of players. Other orderings of the players' names might be done by batting average, age, or height. The information that determines the ordering is called the key. We would expect that each key would give a different permutation of the names. If there are twenty-five players on the team, there are different permutations of the players.

This number of permutations is huge. In fact it is 15511210043330985984000000, but writing it like this isn't all that instructive, while leaving it as a product as we originally had makes it easier to see where the number comes from. We just need to find a more compact way of writing these products.

Definition 2.2.5 Factorial. If n is a positive integer then n factorial is the product of the first n positive integers and is denoted $n!$. Additionally, we define zero factorial, $0!$, to be 1.

$$n! = n(n - 1)(n - 2) \cdots (2)(1),$$
 the product of integers from 1 through n .

Notice that $n! = n \cdot (n - 1)!$

So if there are n different objects, there are $n!$ permutations of those those objects. No object is repeated in any particular arrangement.

A more general situation is when you have n objects, and you need to pick k object out of those and arrange them. In this situation, there are again n possible choices for the first position, $n - 1$ choices for the second, going down to $n - k + 1$ choices for the k^{th} position. By the multiplication rule, there are $n * (n - 1) * (n - 2) \dots (n - k + 1)$ ways to do this. This number, which denotes the number of distinct permutations of n objects taken k at a time, is denoted by $P(n, k)$:

$$P(n, k) = n \cdot (n - 1) \cdot (n - 2) \cdot \dots \cdot (n - k + 1) = \prod_{j=0}^{k-1} (n - j) = \frac{n!}{(n - k)!}.$$

Example: A club has eight members eligible to serve as president, vice-president, and treasurer. How many ways are there of choosing these officers?

Solution 1: Using the rule of products. There are eight possible choices for the presidency, seven for the vice-presidency, and six for the office of treasurer. By the rule of products there are ways of choosing these officers.

Solution 2: Using the permutation formula. We want the total number of permutations of eight objects taken three at a time:

$$P(8, 3) = 8 \cdot 7 \cdot 6 = \frac{8!}{(8 - 3)!} = 336$$

Example: Consider only the digits 1, 2, 3, 4, and 5.

1. How many three-digit numbers can be formed if no repetition of digits can occur?
2. How many three-digit numbers can be formed if repetition of digits is allowed?
3. How many three-digit numbers can be formed if only non-consecutive repetition of digits are allowed?

1. This is permutation of 1, 2, 3, 4, 5 taken 3 at a time, which is $P(5, 3) = 5 \cdot 4 \cdot 3 = 60$. Alternatively, the first position in a 3-digit number can be filled with any one of the 5 numbers, the second position with any one of the remaining 4 numbers, and the last position by any of the remaining 3 numbers, for a total of $5 \cdot 4 \cdot 3 = 60$ possible numbers.

2. Each of the 3 positions can be filled with any one of the 5 digits, so that you can make $5*5*5=125$ different numbers.

3. The first position can be filled in 5 ways. The second position has 4 choices, since it can not be the number chosen for the first position. The third position can be any of the 5 numbers, except the one chosen for second position, so you have 4 choices. The total possibilities are $5*4*4=80$ if non-consecutive repetitions are allowed.